1.INTERACTIVE APPLET TO PRACTISE EXPONENTIAL FORM

Aim

To create an interactive GeoGebra applet for practising exponential form.

Procedure

Step 1: Open GeoGebra

- Open GeoGebra Classic and choose a view with the Input Bar and tools menu.

Step 2: Create variables for Question 1

- Enter these commands in the Input Bar:

```
base1=9
exp1=3
question1=base1^exp1
```

- This gives $9^3 = 729$.

Step 3: Display Question 1

- Choose the Text tool and display a question asking students to identify the base and exponent for the given number.
- For this click text option from tools menu and enable LaTeX Formula option for the mathematical expression. And enter question by below command

```
\text{Find the exponent and base of}:question1
```

- For a dynamic question, insert the current GeoGebra object question1 into the text object so it updates when their values change.

Step 4: Create input boxes for Question 1

- Create two answer variables:

```
ans1=0
ans2=0
```

- Create the Input Box from tools menu. In input dialog box set the caption to Base and link it to ans1. Similarly create another input box set the caption to Exponent and link it to ans2.

Step 5: Create the first Check button

- Create Boolean variable:

```
check1=false
```

- Create a button with caption Check. In script area, enter:

```
SetValue(check1,true)
```

- Click OK.

Step 6: Add feedback for Question 1

- Enter the following command in the Input Bar to display feedback only after the Check button is clicked:

```
feedback1=If(check1,If(ans1^ans2==question1,"correct!","Wrong!"),"")
```

- The entered base and exponent are correct when ans1^ans2 equals question1.

Step 7: Create variables for Question 2

- Enter these commands:

```
Base2=10
Exp2=3
question2=base2^exp2
```

- This gives $10^3 = 1000$.

Step 8: Display Question 2

- Choose the Text tool and display a question asking students to identify the base and exponent for the given number.
- For this click text option from tools menu and enable LaTeX Formula option for the mathematical expression. And enter question by below command

```
\text{Find the exponent of the number}:question2\times\text{with base}base2
```

- For a dynamic question, insert the current GeoGebra object question2 and base2 into the text object so it updates when their values change
- Click OK

Step 9: Create the answer input for Question 2

- Create answer variables:

```
ans3=0
```

- Create the Input Box from tools menu. In input dialog box set the caption to Exponent and link it to ans3.

Step 10: Create the second Check button

- Create Boolean variable:

```
Check2=false
```

- Create a button with caption Check. In script area, enter:

```
SetValue(check2,true)
```

- Click OK.

Step 11: Add feedback for Question 2

- Enter the following command in the Input Bar to display feedback only after the Check button is clicked:

```
feedback2=If(check2,If(ans3==exp2,"correct!","Wrong!"),"")
```

- If ans3 equals exp2, the applet displays Correct! Otherwise, it displays Wrong! Feedback remains blank until Check is clicked.

Step 12: Create the Next Question button

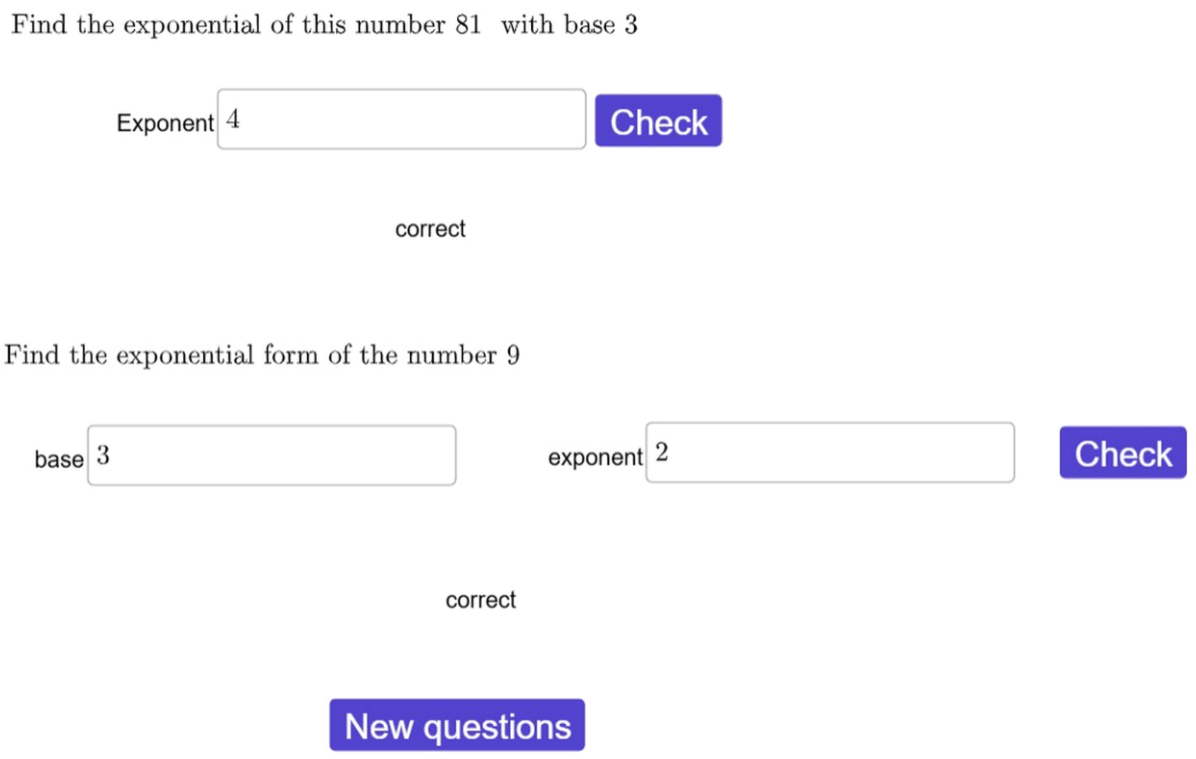
- Create a button captioned Next Question. Open its On Click scripting area and paste this complete script. The random ranges below are example choices for positive integer practice questions.

```
SetValue(base1,RandomBetween(1,10))
SetValue(exp1,RandomBetween(1,10))
SetValue(base2,RandomBetween(1,10))
SetValue(base1,RandomBetween(1,10))
SetValue(ans1,0)
SetValue(ans2,0)
SetValue(ans3,0)
SetValue(check1,false)
SetValue(check2,false)
```

- Click OK.
- The button generates new values, resets the answer inputs to zero, and clears feedback by resetting check1 and check2 to false. Make sure the displayed question text is dynamic and linked to the current GeoGebra objects.

Result

An interactive GeoGebra applet is created with two types of exponential-form questions, input boxes, Check buttons, feedback messages, and a Next Question button.



2. CREATING A GEOGEBRA APPLET TO DEMONSTRATE THE LAWS OF EXPONENTS

Aim

To create an interactive GeoGebra applet to demonstrate the product rule, quotient rule, and power rule of exponents when the base is the same, using sliders and practice questions with answer checking.

Procedure

Step 1: Create the Heading

- Open GeoGebra Calculator.
- Select the Text tool from the toolbar.
- Enter the heading "**Exponents with the Same Base but Different Powers**." Using text option.
- Click OK and place the heading in the desired position in the Graphics View.

Step 2: Create Sliders for the Base and Exponents

- Open the Measures tools menu and select the Slider tool.
- Click on the desired position in the Graphics View.
- In the slider dialog box, enter the name a.
- Set the minimum and maximum values according to your requirements and set the step value to 1.
- Click OK to create the slider.
- Repeat the same procedure to create two more sliders named n and m.
- The sliders represent the base a and exponents n and m.

Step 3: Create Variables for the Exponent Operations

- Select the Input Bar and create variables named add, sub, and multiply.
- Enter the following commands:

```
add = n + m
sub = n - m
multiply = n * m
```

The variable add stores the sum of the exponents, sub stores the difference, and multiply stores their product.

Step 4: Demonstrate the Product Rule

- Select the Text tool and choose the desired position in the Graphics View.
- Select the LaTeX Formula option and enter the product rule expression:

```
\text{\bf product rule :} a^{m}\ times a^{n}= a^{m+n}= a ^{add}
```

- Insert the dynamic variables a, n, m, and add using the appropriate GeoGebra text-object options.
- Click OK. The expression updates automatically when the sliders change.

Step 5: Demonstrate the Quotient Rule

- Select the Text tool and choose the desired position.
- Select the LaTeX Formula option and enter the quotient rule expression:

```
\text{\bf Quotient rule :} \frac{a^{m}}{a^{n}}= a^{m-n}= a ^{sub}
```

- The \frac command to display the fraction correctly.
- Insert the dynamic variables a, n, m, and sub.
- Click OK. The expression updates automatically when the sliders change.

Note: The quotient rule requires $a \neq 0$.

Step 6: Demonstrate the Power Rule

- Select the Text tool and choose the desired position.
- Select the LaTeX Formula option and enter the power rule expression:

```
\text{\bf Power rule :} ({a^{m}})^{n}=a^{m*n}=a ^{multiply}
```

- Insert the dynamic variables a, n, m, and multiply.
- Click OK. The expression updates automatically when the sliders change.

Step 7: Test the Interactive Expressions

- Move the base slider a and observe the changes.
- Move the exponent sliders n and m.
- Verify that the product rule adds the exponents.
- Verify that the quotient rule subtracts the exponents.
- Verify that the power rule multiplies the exponents.

For example, set a = 6, n = 7, and m = 5:

- Product rule: $6^7 \times 6^5 = 6^{12}$
- Quotient rule: $6^7 / 6^5 = 6^2$
- Power rule: $(6^7)^5 = 6^{35}$

Step 8: Create Variables for the Practice Questions

1. Select the Input Bar and create the variables for the three practice questions.
2. For Question 1, use b1, m1, and n1. For Question 2, use b2, m2, and n2. For Question 3, use b3, m3, and n3.
3. Enter variables with below values by using text option;

```
b1 = 3
m1 = 5
n1 = 8
b2 = 2
m2 = 5
n2 = 3
b3 = 6
```

```
m3 = 4
n3 = 3
```

Step 9: Display the Practice Questions

- Select the Text tool and enter the heading “Solve the following.”
- Now double tab the above text (solve the following).
- Now you will get text dialog box, enter the questions by below format;

```
\text{solve the following }\\
1.b1^{m1}\times b1^{n1}\\
2.\frac{b2^{m2}}{b2^{n2}}\\
3.(b3^{m3})^{n3}
```

- Click OK after creating each text object and arrange the questions neatly in the Graphics View.
- Display the following expressions:

Solve the following

1. $3^5 \times 3^8$
2. $2^5 / 2^3$
3. $(6^4)^3$

Step 10: Create the Answer Input Boxes

- Create three dummy variables named ans1, ans2, and ans3, each with initial value zero.

```
ans1 = 0
ans2 = 0
ans3 = 0
```

- Select the Input Box tool.
- Create an input box captioned Answer 1 and link it to ans1.
- Create Answer 2 linked to ans2 and Answer 3 linked to ans3.
- Click OK for each input box.

Students can enter the final exponent of each simplified expression.

Step 11: Create the Check Buttons

- Create Boolean variables check1, check2, and check3 and set each to false.

```
check1 = false
check2 = false
check3 = false
```

- Select the Button tool and create a button captioned Check 1.
- In its click-script box, enter:

```
SetValue(check1, true)
```

- Similarly, Create Check 2 and Check 3 with these commands:

```
SetValue(check2, true)
SetValue(check3, true)
```

Clicking a Check button changes its corresponding Boolean variable from false to true.

Step 12: Display Feedback for the Practice Questions

- Create a feedback text object for each question and use the following commands: By using text option enter the below code:

```
feedback1 = If(check1, If(ans1 == m1 + n1, "Correct!", "Wrong!"), "")
feedback2 = If(check2, If(ans2 == m2 - n2, "Correct!", "Wrong!"), "")
feedback3 = If(check3, If(ans3 == m3 * n3, "Correct!", "Wrong!"), "")
```

- Ensure that each text object displays the value of its corresponding feedback variable.
- Place each feedback object near its question.

The feedback compares ans1 with the sum of the exponents, ans2 with their difference, and ans3 with their product. If the corresponding Check button has not been clicked, no feedback is displayed.

Expected answers for the initial questions:

- Question 1: 5 + 8 = 13
- Question 2: 5 − 3 = 2
- Question 3: 4 × 3 = 12

Step 13: Create the Next Question Button

- Select the Button tool and enter the caption Next Question.
- Open the scripting section and enter these commands in the click-script box:

```
SetValue(b1, RandomBetween(2, 9))
SetValue(m1, RandomBetween(1, 9))
SetValue(n1, RandomBetween(1, 9))

SetValue(b2, RandomBetween(2, 9))
SetValue(m2, RandomBetween(2, 9))
SetValue(n2, RandomBetween(1, m2 - 1))

SetValue(b3, RandomBetween(2, 9))
SetValue(m3, RandomBetween(1, 9))
SetValue(n3, RandomBetween(1, 9))

SetValue(ans1, 0)
SetValue(ans2, 0)
SetValue(ans3, 0)

SetValue(check1, false)
```

```
SetValue(check2, false)
SetValue(check3, false)
```

- Click OK to create the button.
- When clicked, new bases and exponents are generated, answer variables reset to zero, check variables reset to false, and previous feedback disappears.
- Ensure the question text objects use b1, m1, n1, b2, m2, n2, b3, m3, and n3 so that questions update dynamically.

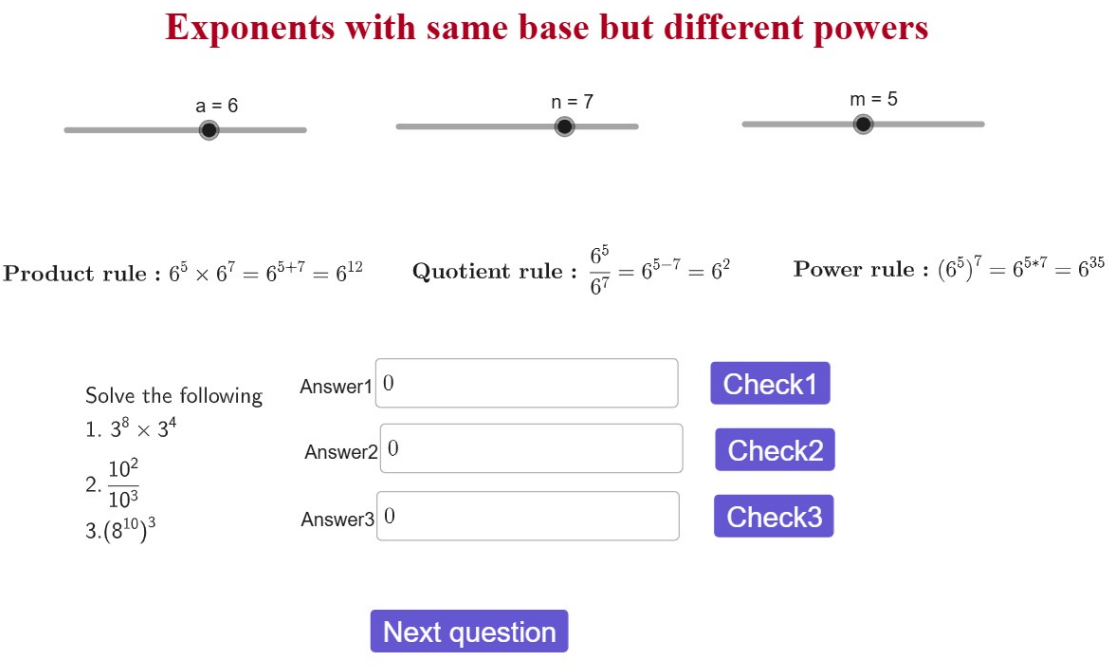
Important: Use dynamic LaTeX text objects linked to the variables rather than fixed expressions if you want the questions to change when Next Question is clicked.

Step 14: Test the Applet

- Move the sliders and verify that all three exponent laws update correctly.
- Enter the expected answer for each practice question.
- Click the corresponding Check button and verify that Correct! is displayed.
- Enter an incorrect answer and verify that Wrong! is displayed.
- Click Next Question and confirm that new questions appear.
- Verify that the answer boxes reset and previous feedback disappears.
- Repeat the process to ensure that the applet functions correctly.

Result

Thus, an interactive GeoGebra applet was created to demonstrate the product rule, quotient rule, and power rule of exponents using sliders and dynamic mathematical expressions. Practice questions, answer input boxes, Check buttons, and a Next Question button were also included.



3.CREATING A GEOGEBRA APPLET TO FIND THE UNIT DIGIT OF EXPONENTIAL NUMBERS

Aim

To create an interactive GeoGebra applet that generates random exponential questions, allows students to enter their answers, checks the answers, provides feedback, and generates new questions for repeated practice.

Procedure

Step 1: Insert the Rules for Finding the Unit Digit

- Open GeoGebra Calculator.
- Copy the image containing the rules for finding the unit digit of exponential expressions.
- Paste the image into the GeoGebra applet.

Step 2: Create Variables for the Base and Exponent

- Identify the base and exponent from the Class 7 Term 2 textbook, Example 3.11, Question 1.
- In this example, the base is 25 and the exponent is 23.
- Enter the following commands in the Input Bar:

```
x = 25
y = 23
```

- Here, x represents the base and y represents the exponent. These variables will be used to display the exponential expression and generate new questions.

Step 3: Display the Exponential Question

- Select the Text tool from the toolbar.
- Open the Text dialog box by double-clicking the text object.
- Select the LaTeX option.
- Enter \text{} in the text field.
- Click the plus symbol and select Empty Object.
- Select the variable x to insert the base dynamically.
- Insert the variable y as the exponent.
- Click OK to display the exponential expression.
- When the values of x and y change, the displayed question will update automatically.

Step 4: Create the Answer Input Box

- Create a variable named ans and assign it the initial value zero.
- Enter the following command in the Input Bar:

```
ans = 0
```

- Select the Input Box tool from the toolbar.
- Enter the caption as Answer.
- Link the input box to the object ans.
- Click OK to create the input box.
- Students can enter their answers directly into this box.

Step 5: Calculate the Last Digit of the Base

- Create a variable named LastDigit.
- Enter the following command in the Input Bar:

```
LastDigit = Mod(x, 10)
```

- The Mod command calculates the remainder when the base is divided by 10.
- This gives the last digit of the base. For example, if x = 25, then LastDigit = 5.
- This variable is used to identify the repeating cycle of unit digits.

Step 6: Calculate the Cycle Length

- Create a variable named CycleLength.
- Enter the following command in the Input Bar:

```
CycleLength = If(LastDigit == 0 || LastDigit == 1 || LastDigit == 5 || LastDigit == 6, 1, If(LastDigit == 4 || LastDigit == 9, 2, 4))
```

- This command determines the cycle length according to the last digit of the base.
- Last digits 0, 1, 5, and 6 have a cycle length of 1.
- Last digits 4 and 9 have a cycle length of 2.
- Last digits 2, 3, 7, and 8 have a cycle length of 4.
- GeoGebra automatically assigns the appropriate cycle length based on LastDigit.

Step 7: Calculate the Position in the Cycle

- Create a variable named Position.
- Enter the following command in the Input Bar:

```
Position = Mod(y - 1, CycleLength) + 1
```

- The Mod command calculates the remainder.
- Subtracting one from the exponent and adding one to the remainder helps identify the correct position in the repeating cycle.
- This formula also works when the exponent is exactly divisible by the cycle length because it selects the last element of the cycle.

Step 8: Calculate the Correct Answer

- Create a variable named CorrectAnswer.
- Enter the following command in the Input Bar:

CorrectAnswer = If(LastDigit == 0, 0, If(LastDigit == 1, 1, If(LastDigit == 2, Element({2, 4, 8, 6}, Position), If(LastDigit == 3, Element({3, 9, 7, 1}, Position), If(LastDigit == 4, Element({4, 6}, Position), If(LastDigit == 5, 5, If(LastDigit == 6, 6, If(LastDigit == 7, Element({7, 9, 3, 1}, Position), If(LastDigit == 8, Element({8, 4, 2, 6}, Position), If(LastDigit == 9, Element({9, 1}, Position)))))))))

- This command checks the last digit of the base and selects the corresponding repeating cycle.
- The Element command selects the unit digit at the position calculated in Step 7.
- The resulting unit digit is stored in CorrectAnswer.
- This method finds the unit digit without calculating the entire exponential expression.

Step 9: Create the Check Button

- Create a Boolean variable named checked and assign it the initial value false.
- Enter the following command in the Input Bar:

checked = false

- Select the Button tool from the toolbar.
- Enter the caption as Check.
- Open the scripting section of the button.
- In the appropriate click-script box, enter the following command:

SetValue(checked, true)

- Click OK to create the button.
- When students click the Check button, the value of checked changes from false to true, allowing the applet to display feedback.

Step 10: Display Feedback for the Answer

- Create a text object to display the result of the student's answer.
- Enter the following conditional command:

feedback = If(checked, If(ans == CorrectAnswer, "Correct! 🎉", "Wrong! ❌"), "")

- The command checks whether the Check button has been clicked.
- If checked is true, GeoGebra compares ans with CorrectAnswer.
- If both values are equal, the applet displays Correct! 🎉.
- If the values are different, the applet displays Wrong! ❌.
- If the Check button has not been clicked, no feedback is displayed.

Step 11: Create the Next Question Button

- Select the Button tool from the toolbar.
- Enter the caption as Next Question.
- Open the scripting section of the button.
- Enter the following commands in the script box:

SetValue(x, RandomBetween(2, 99))
SetValue(y, RandomBetween(2, 99))

SetValue(ans, 0)
SetValue(checked, false)

- Click OK to create the button.
- When the student clicks Next Question, a new random base and exponent are generated.
- The answer variable is reset to zero and checked is reset to false.
- The previous feedback disappears and the new exponential question is displayed automatically.
- The variables LastDigit, CycleLength, Position, and CorrectAnswer update automatically when the base or exponent changes.

Step 12: Test the Applet

- Run the applet and observe the initial question.
- Enter the correct unit digit in the Answer input box.
- Click Check and verify that the applet displays the correct feedback.
- Enter an incorrect answer and check whether the applet displays the wrong-answer message.
- Click Next Question and verify that a new question appears.
- Repeat the process to ensure that all components work correctly.

Result

Thus, an interactive GeoGebra applet was created to display the rules for finding unit digits, generate random exponential questions, accept students' answers, calculate the correct unit digit, check the answers, display feedback, and generate new questions for repeated practice.

